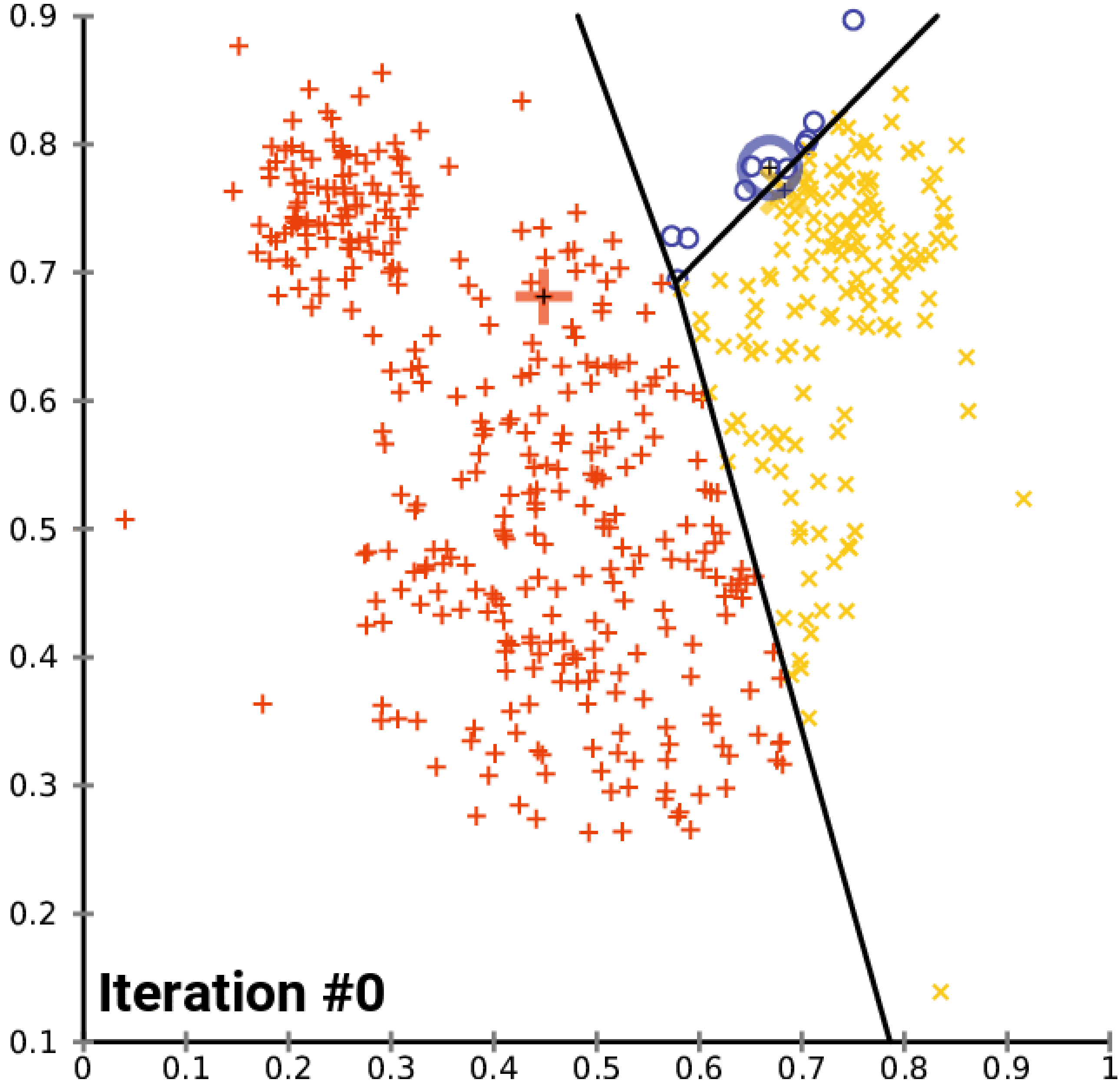
K-Means Clustering



What is K-Means Clustering?

K-Means Clustering is an **unsupervised machine learning algorithm** used to group similar data points into clusters.

It divides the dataset into **K clusters** based on similarity.

Main Idea of K-Means

The algorithm:

1. Chooses K centroids
2. Assigns data points to nearest centroid
3. Updates centroids
4. Repeats until clusters stabilize

Important Terms

Term	Meaning
Cluster	Group of similar data points
Centroid	Center point of a cluster
K	Number of clusters

Distance Formula

K-Means usually uses **Euclidean Distance**.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Where:

Symbol	Meaning
(x_1, y_1)	First point coordinates
(x_2, y_2)	Second point coordinates
(d)	Distance between points

Working of K-Means

Steps

1. Select number of clusters (K)
2. Initialize centroids randomly
3. Assign points to nearest centroid
4. Recalculate centroids
5. Repeat until no changes occur

Example of K-Means Clustering

Cluster Example

Suppose we have customer data:

Customer	Spending Score
A	High
B	High
C	Low
D	Low

If ($K = 2$):

- Cluster 1 → High spenders
- Cluster 2 → Low spenders

Choosing the Value of K

Elbow Method

Used to find the optimal number of clusters.

The graph looks like an elbow shape.

Objective Function

K-Means minimizes:

$$J = \sum_{i=1}^{k} \sum_{x_j \in C_i} ||x_j - \mu_i||^2$$

Where:

Symbol	**Meaning**
(J)	Cost function
(C_i)	Cluster
(\mu_i)	Centroid

Types of Clustering

1. Hard Clustering

Each point belongs to only one cluster.

Example:

- K-Means

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2. Soft Clustering

Points can belong to multiple clusters.

Example:

- Fuzzy C-Means

Advantages of K-Means

- Simple and easy to implement
- Fast for large datasets
- Efficient clustering

- Works well with numerical data

Disadvantages of K-Means

- Need to choose K manually
- Sensitive to outliers
- Different initial centroids may give different results
- Works poorly with non-spherical clusters

Applications of K-Means

- Customer Segmentation
- Image Compression
- Recommendation Systems
- Market Analysis
- Document Clustering

Comparison with Other Algorithms

Algorithm	Type	Speed	Best For
K-Means	Unsupervised	Fast	Clustering
SVM	Supervised	Medium	Classification
Naive Bayes	Supervised	Very Fast	Text data
Decision Tree	Supervised	Medium	Rule-based problems

Python Example

```python
from sklearn.cluster import KMeans
import numpy as np

# Data
X = np.array([
    [1, 2],
    [1, 4],
    [1, 0],
    [10, 2],
    [10, 4],
    [10, 0]
])

# Create model
kmeans = KMeans(n_clusters=2)

# Train model
kmeans.fit(X)

# Cluster labels
print(kmeans.labels_)
```

Important Interview Questions

1. What is K-Means?

An unsupervised algorithm used for clustering.

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2. What is a Centroid?

The center point of a cluster.

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3. What is the Elbow Method?

A method to determine the optimal value of K.

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4. Is K-Means supervised or unsupervised?

Unsupervised.

Short Notes

Definition

K-Means is an unsupervised machine learning algorithm used to divide data into K clusters.

Main Goal

Group similar data points together.

Advantages

- Fast
- Simple
- Efficient

Disadvantages

- Sensitive to outliers
- Need to choose K manually

Applications

- Customer segmentation
- Image compression
- Market analysis
- Recommendation systems